

LiDAR – Project

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Project Overview

Goal: To utilize a sensor and its data to generate several type of visualizations

- Low-cost, yet effective data acquiring sensors.
- World is becoming more electrified and tech-oriented, meaning lots of data involved
- Lidar sensors - great for object detection
- MCU to process and transmit data

Device: ESP32-S3-DevKit-C1



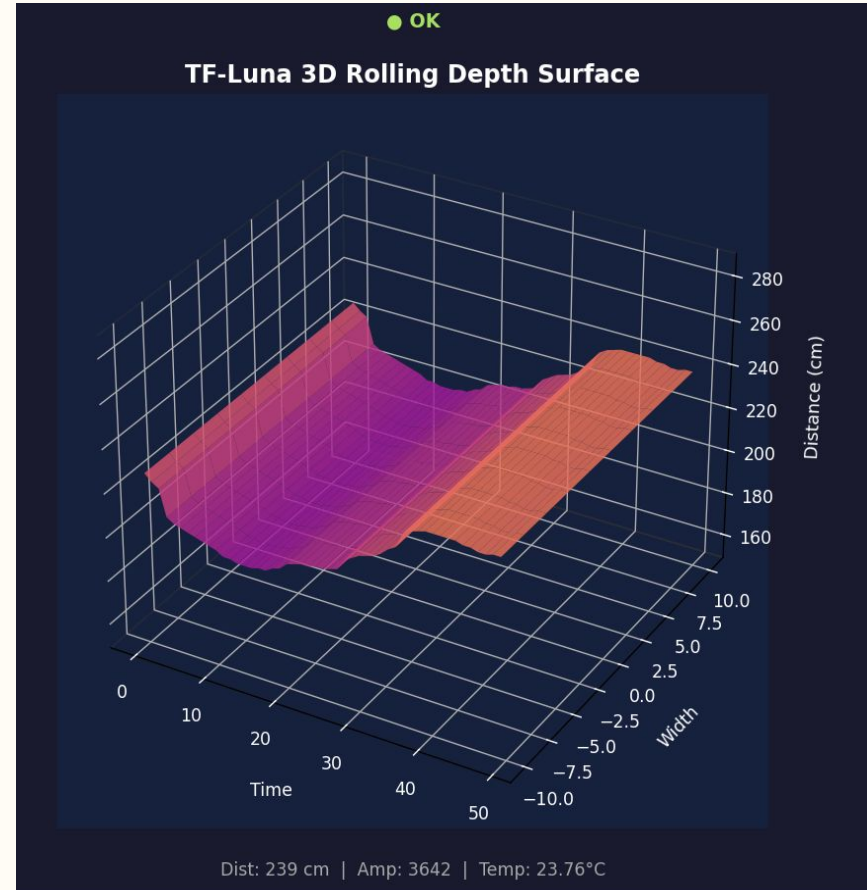
System Architecture

Hardware-wise

- ESP32 MCU

In regards to programming, I went with both C++ and Python

- C++: good for receiving and processing hundreds, if not, thousands of raw sensor data coming in.
 - Sent through the UART port on the MCU
 - Communicates with the GPIOs on the MCU
- Python: imported serial data through the UART port within the ESP32 and assist in creating visualizations
 - Reflectance and other information from objects translated into data points



Interfaces and Peripherals

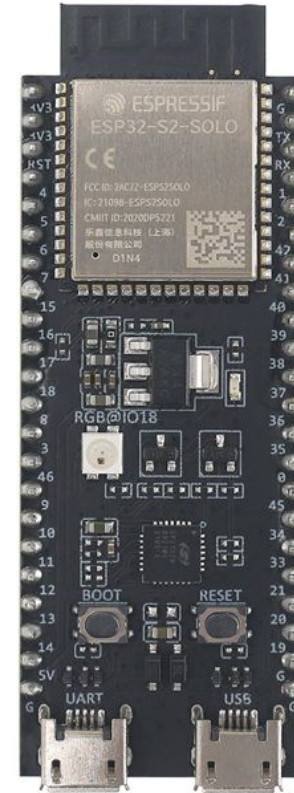
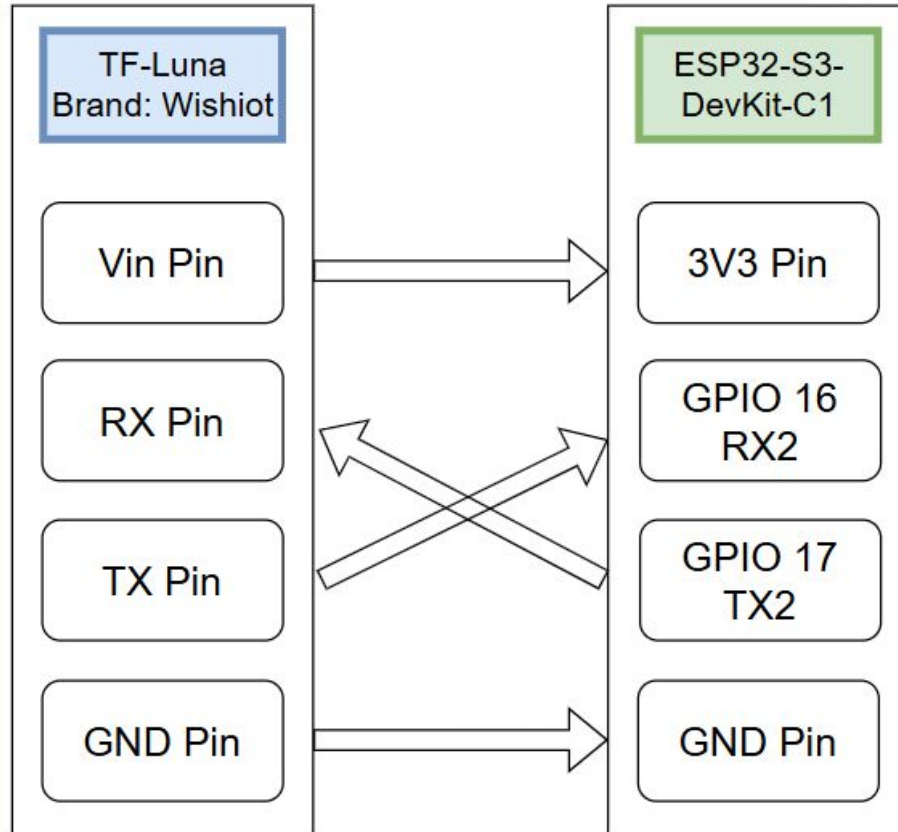
The ESP32-S3-DevKit-C1 Microcontroller was utilized for this project

No other peripherals or ICs were used, other than the sensor and the MCU.

There were plans to incorporate an OLED screen as well and servo motors, but not implemented due to lack of time

Pin	Components
GPIO 16	ESP32-S3-DevKitC1
GPIO 17	TF-Luna Lidar Sensor
3V3	USB-C
GND	Laptop

Interfaces and Peripherals



Verification and Testing

Testing and verification strategy composed in two ways:

Hardware:

- Checking that the RGB leds are on within the MCU. It would verify that the MCU is receiving power since it has no separate battery source of its own.
- And checking that the IR sensor is activated with the lidar sensor upon receiving power through the MCU

Software:

- Verifying through the serial monitor terminal that the outputs or graphs are displaying correctly within the correct COM port and appropriate baud rate



Project Demonstration

Live Demo or Link

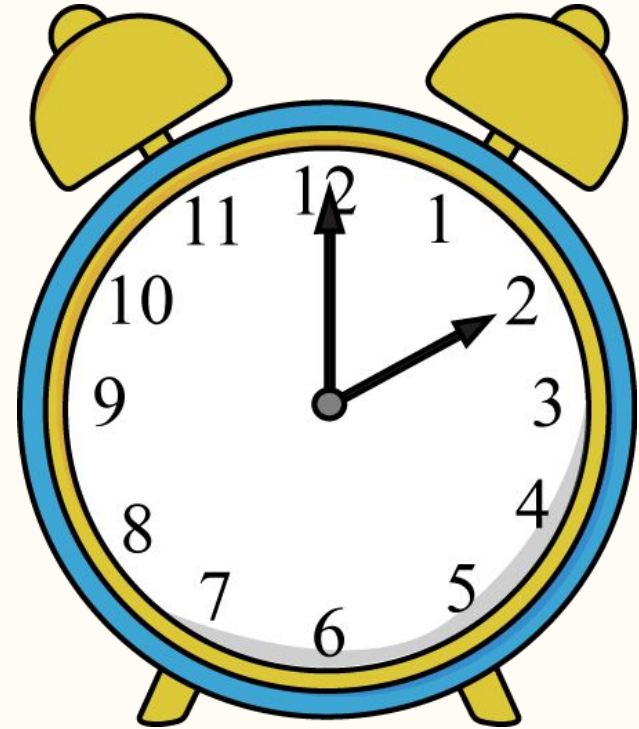
Conclusion

In summary:

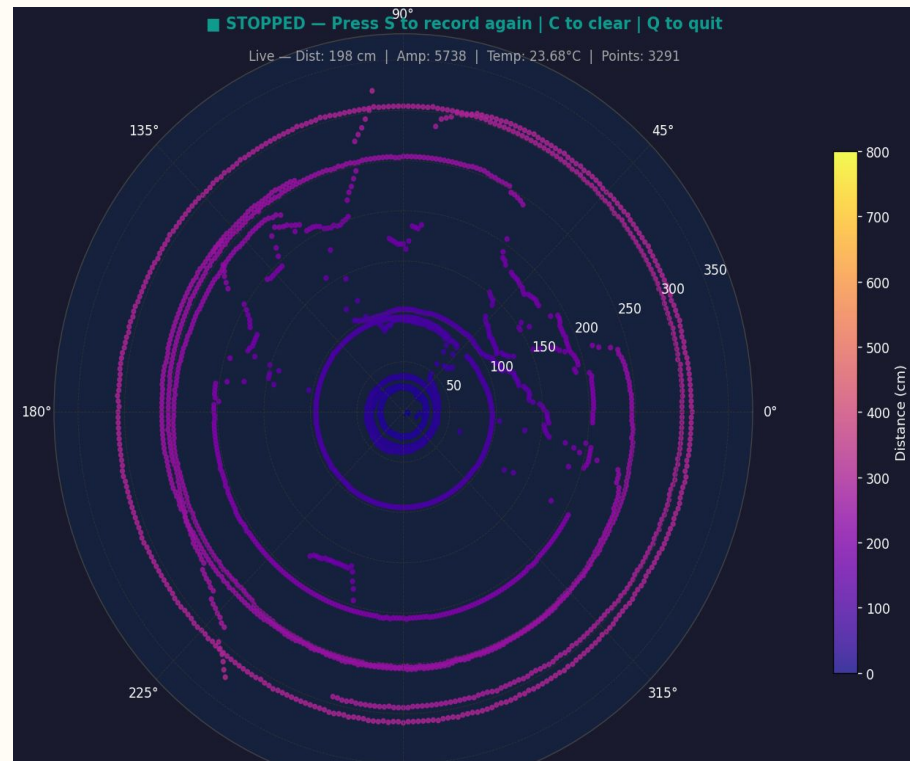
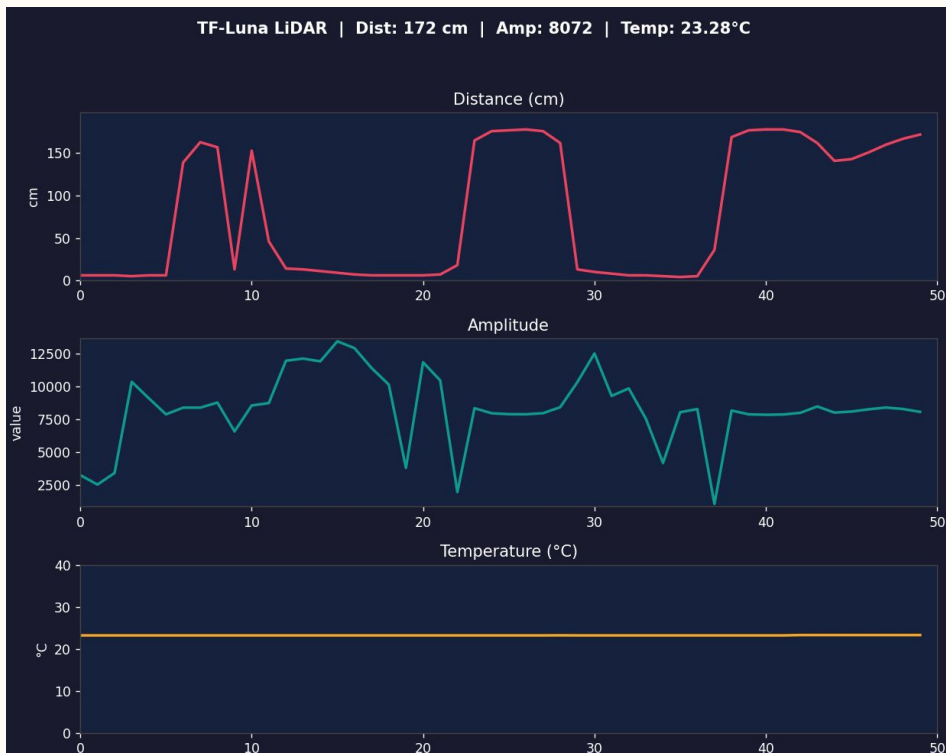
- The TF-Luna lidar sensor was successfully powered on and outputting data that would be visualized onto different display graphs.
- It shows how just a sensor worth of data can be used to display different forms of graphs or even applications as the user needs, if the equipment or tools are available.

Challenges/Issues faced:

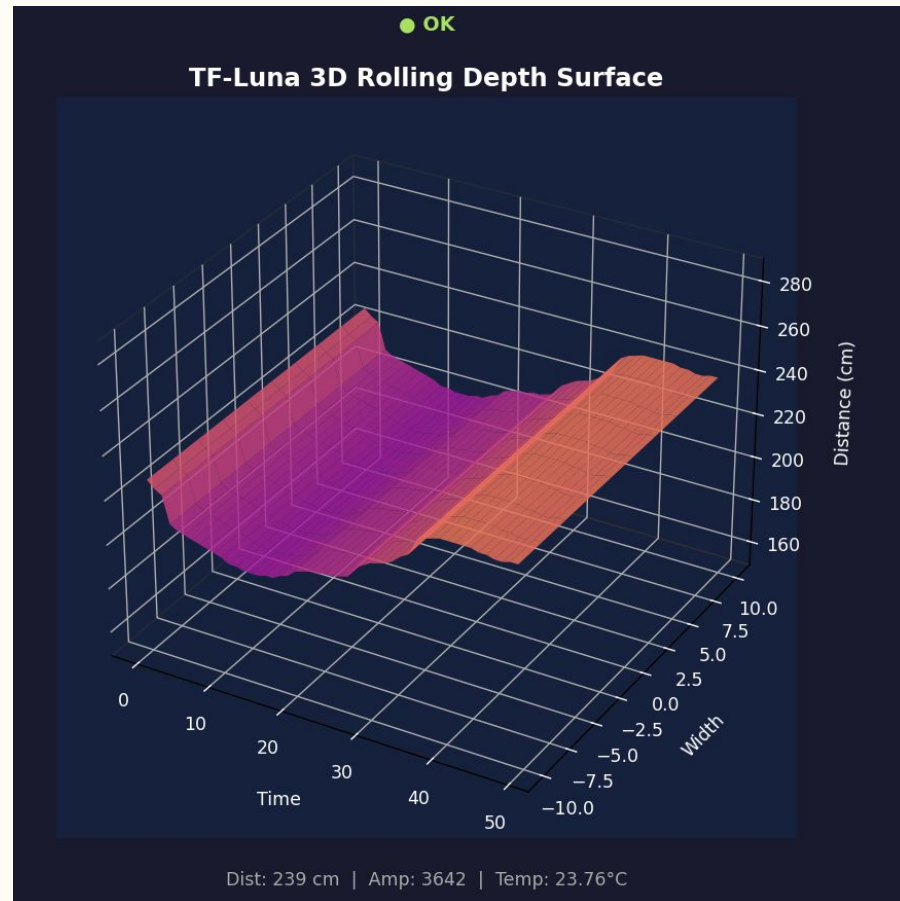
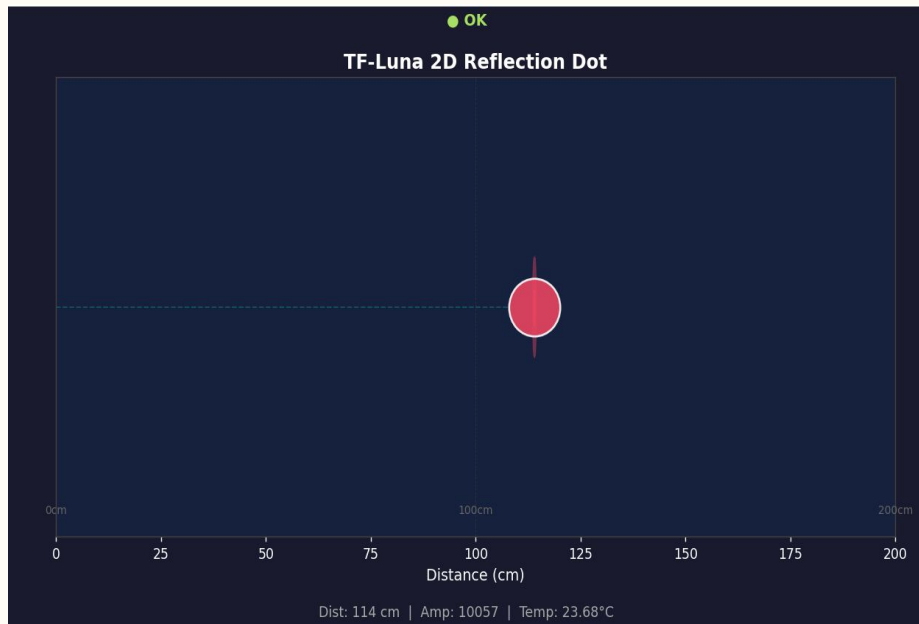
- Time management - Balancing and focus was a big conflict with outside duties.
- Figuring out the pinout for the lidar sensor since pins are not consistent among brands



Conclusion



Conclusion

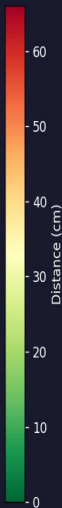


Conclusion

⚠️ TOO CLOSE

TF-Luna Proximity Heatmap

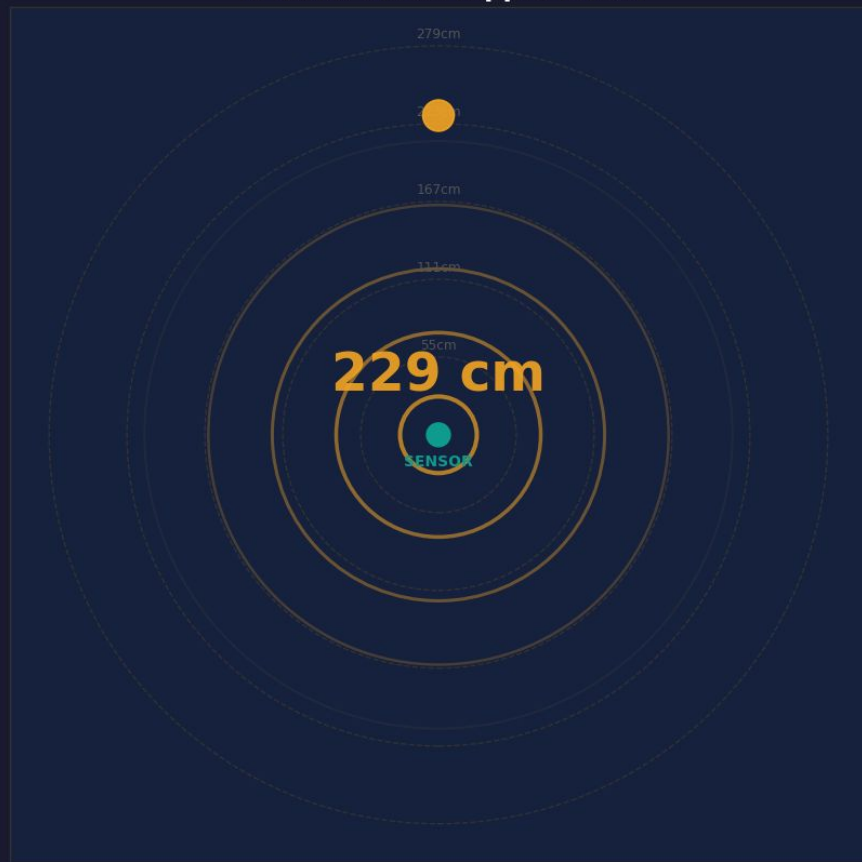
15 cm



Dist: 15 cm | Amp: 2463 | Temp: 23.76°C

● OK

TF-Luna Sonar Ripple Effect



Dist: 229 cm | Amp: 4278 | Temp: 23.84°C | Range: 279 cm

Conclusion

